



LoRA Synergy™

Stop guessing which LoRA goes with which checkpoint. Let SmartGallery tell you.

LoRA Synergy is a zero-API, fully offline LoRA matchmaker and injector built into SmartGallery's **Remix Workflow** → **Nodepad**. It scans the actual `.safetensors` files on your disk, reads their metadata, and tells you *exactly* which of your installed LoRAs are compatible with the checkpoint currently loaded in your workflow — then wires everything up for you in one click.

No more scrolling through 400 LoRAs guessing if they're SDXL, Flux, or SD1.5. No more black-screen renders because you mixed architectures. No more digging through CivitAI tabs to remember the trigger word.

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Why LoRA Synergy exists

If you've used ComfyUI for more than a week, you know this pain:

- You load someone else's workflow, want to try a different LoRA, and have no idea which of your 200 files will even work with that checkpoint.
- You attach a LoRA, hit queue, and get a cryptic tensor mismatch error five minutes later.
- You *do* find a compatible LoRA, but forgot the trigger word, so your generation looks nothing like the preview.
- Wiring a new `LoraLoader` node into an existing graph means manually rerouting `MODEL` and `CLIP` outputs through every downstream node — tedious and error-prone.

LoRA Synergy solves all four problems in a single panel, without ever calling an external API or requiring an internet connection.

What it actually does

1. **Reads your graph** and detects every valid injection target — checkpoint loaders, UNET loaders, GGUF loaders, and existing LoRA nodes — directly from the currently open workflow in the Nodepad.

2. **Interrogates your local LoRA files** — not a database, not the cloud, your actual files on disk — to determine their base architecture (SD1.5, SDXL, Flux, and more) by inspecting the safetensors header.
3. **Cross-references** that architecture against the checkpoint you have loaded and instantly buckets every LoRA in your library into ranked compatibility groups.
4. **Surfaces trigger words** automatically, pulling from CivitAI metadata (if available) and from SmartGallery's own history of files you've previously generated.
5. **Auto-wires the graph**: adds the new `LoraLoader` (or `LoraLoaderModelOnly`) node and rewires every downstream `MODEL` / `CLIP` connection that used to point at your checkpoint — so nothing breaks.

All of this happens **locally**. LoRA Synergy never uploads your files or workflow anywhere.

Quick Start

1. Open **Remix Workflow** on any media file (or from the Tools menu / Library).
2. Switch to the `{ }` **Nodepad** tab.
3. Make sure you're in **API Format** (LoRA Synergy currently requires API format — the toolbar will prompt you to switch if needed).
4. Click  **LoRA Synergy™** in the toolbar.
5. Pick your **Injection Target** (your checkpoint, UNET, or an existing LoRA node).
6. Search or browse the ranked LoRA list —  Perfect Matches are shown first.
7. Click a LoRA, copy a trigger word chip if one appears, adjust strength if needed.
8. Click  **Execute Synergy**.

That's it — the node is created and wired, and you're ready to queue.

Understanding the Match Categories

When you open the injector, your entire local LoRA library is automatically sorted into four groups so you never waste time on incompatible files:

Category	What it means
 Perfect Match	Verified directly from the LoRA's own safetensors header — this is a hard, technical guarantee of architecture compatibility.
 Proven Match	You've successfully used this exact LoRA with this checkpoint (or its architecture) before, according to SmartGallery's history.
 Possible Match	Inferred from the LoRA's filename or the folder it lives in (e.g. a <code>Flux/</code> subfolder) — a good hint, not a guarantee.
 Incompatible	Detected as a different base architecture than your checkpoint. Collapsed by default to keep the list clean, but still one click away if you want to override.

You can always ignore the ranking and search by name — Synergy never blocks you, it just saves you from guessing.

Chain vs. Replace

When your injection target is an existing LoRA node, LoRA Synergy gives you two options:

-  **Add After (Chain)** — creates a brand-new `LoraLoader` node downstream of the one you picked, so you can stack multiple LoRAs.
-  **Replace Existing** — swaps the LoRA in the currently selected node in place, keeping the same node ID and all its existing wiring.

For checkpoint/UNET targets, Synergy always chains a new LoRA node and automatically rewires every node that was previously connected directly to your checkpoint's `MODEL` / `CLIP` outputs — so the graph stays fully connected with zero manual rerouting.

Trigger Words & Memory

Forgetting trigger words is one of the most common reasons a LoRA "doesn't seem to work." LoRA Synergy fixes this two ways:

-  **Official Triggers** — pulled straight from the LoRA's CivitAI metadata when available.
-  **Historical Triggers** — pulled from SmartGallery's own database of prompts you've previously paired with this LoRA.

Click any trigger chip to copy it instantly — and it's automatically saved to the floating  **Memory Triggers** panel, a small draggable widget that keeps every trigger you've copied during your session in one place, ready to paste into your prompt node without re-opening the injector.

Better together: ComfyUI-Lora-Manager

LoRA Synergy works out of the box with **zero configuration** on a standard ComfyUI install. But it reaches its full potential when paired with the excellent [ComfyUI-Lora-Manager](#) by *willmiao*.

If Lora-Manager is installed, LoRA Synergy automatically detects and uses its metadata to unlock:

-  **Preview thumbnails** for every LoRA, right inside the injector panel.
-  **Direct CivitAI links** for the model you're about to inject.
-  **One-click deep link** into the Lora Manager UI itself for further browsing.
- More complete **trigger word** extraction, since Lora-Manager keeps CivitAI metadata cached and up to date locally.

You don't have to install it — Synergy will still find your models by scanning your standard ComfyUI `models/` folder and will still classify architecture compatibility from the safetensors header alone. But if you want the richest, most visual experience, Lora-Manager is the recommended companion.

 github.com/willmiao/ComfyUI-Lora-Manager

Advanced setup (Docker / network drives / custom paths)

By default, SmartGallery finds your models by scanning your ComfyUI `models/` folder. If you run ComfyUI in an isolated Docker container, on a network drive, or use `extra_model_paths.yaml1`, SmartGallery may not be able to see your files automatically.

In that case, set these environment variables before starting the SmartGallery server:

Variable	Purpose
<code>BASE_MODELS_PATH</code>	Use if you moved your entire <code>models/</code> folder elsewhere.
<code>LORAS_PATH</code>	Granular override just for your LoRAs folder.
<code>CHECKPOINTS_PATH</code>	Granular override just for your checkpoints folder.
<code>UNET_PATH</code>	Granular override for FLUX/SD3-style UNET files.

Once set, restart the server — LoRA Synergy will pick up the correct paths automatically.

Step-by-step walkthrough

1. **Open a workflow.** In the gallery, press **B** on any media file that contains a ComfyUI workflow (or open one from your saved **Workflow Library**).
2. **Go to the Nodepad.** Click the `{ }` **Nodepad** tab at the top of the Remix window.
3. **Confirm API Format.** LoRA Synergy only works in API Format — switch the format dropdown in the toolbar if you're in UI Format.
4. **Launch Synergy.** Click  **LoRA Synergy™**. SmartGallery scans your graph for valid targets (checkpoints, UNETs, existing LoRA nodes).
5. **Choose your target.** Pick the node you want to attach a LoRA to from the dropdown. If it's already a LoRA node, choose **Chain** or **Replace**.
6. **Browse ranked results.** Perfect Matches surface first. Use the search box to filter by name or folder if you know what you're looking for.
7. **Preview & inspect.** If Lora-Manager is installed, hover for a thumbnail and click the CivitAI link to double check.
8. **Grab the trigger word.** Click any green or orange trigger chip to copy it — it'll also land in your floating Memory Triggers widget.
9. **Set strengths.** Adjust `Strength Model1` and `Strength CLIP` (CLIP strength is hidden automatically for model-only targets like UNETLoader).
10. **Execute.** Click  **Execute Synergy**. The node is created (or replaced) and every downstream connection is automatically rewired.
11. **Add the trigger to your prompt.** Paste the trigger word you copied into your positive prompt node — Synergy reminds you to do this after every injection.
12. **Queue as usual.** Head back to Auto-Form or My Panel, or queue directly from the Nodepad.

FAQ & Troubleshooting

Why is LoRA Synergy grayed out / asking me to switch format?

It currently only operates on API Format workflows, since it needs to programmatically read and rewrite node connections. Switch the format dropdown in the Nodepad toolbar to *API Format*.

Why does it say "Cannot reach ComfyUI"?

Synergy needs ComfyUI's live node dictionary to know each node's input schema. Make sure ComfyUI is running and that the **ComfyUI Server URL** field at the top of the Remix window points to the right address.

No targets were found in my graph — why?

Synergy looks for checkpoint loaders, UNET/diffusion/GGUF loaders, and existing LoRA loader nodes. If your workflow uses a completely custom loader node type without a recognizable model filename, it won't be detected as a target.

A LoRA I know is compatible shows up under "Incompatible" — what do I do?

Architecture detection is based on the safetensors header and is highly reliable, but not infallible for unusual or mislabeled files. The Incompatible group is collapsed, not hidden — expand it and select it anyway if you're confident.

Does this send my files or workflow anywhere?

No. Every scan, match, and injection happens locally against files already on your machine.

Questions or feedback? Open an issue on the SmartGallery repo — LoRA Synergy is actively evolving.